

# HVL 192V 60AH-F15CL-B

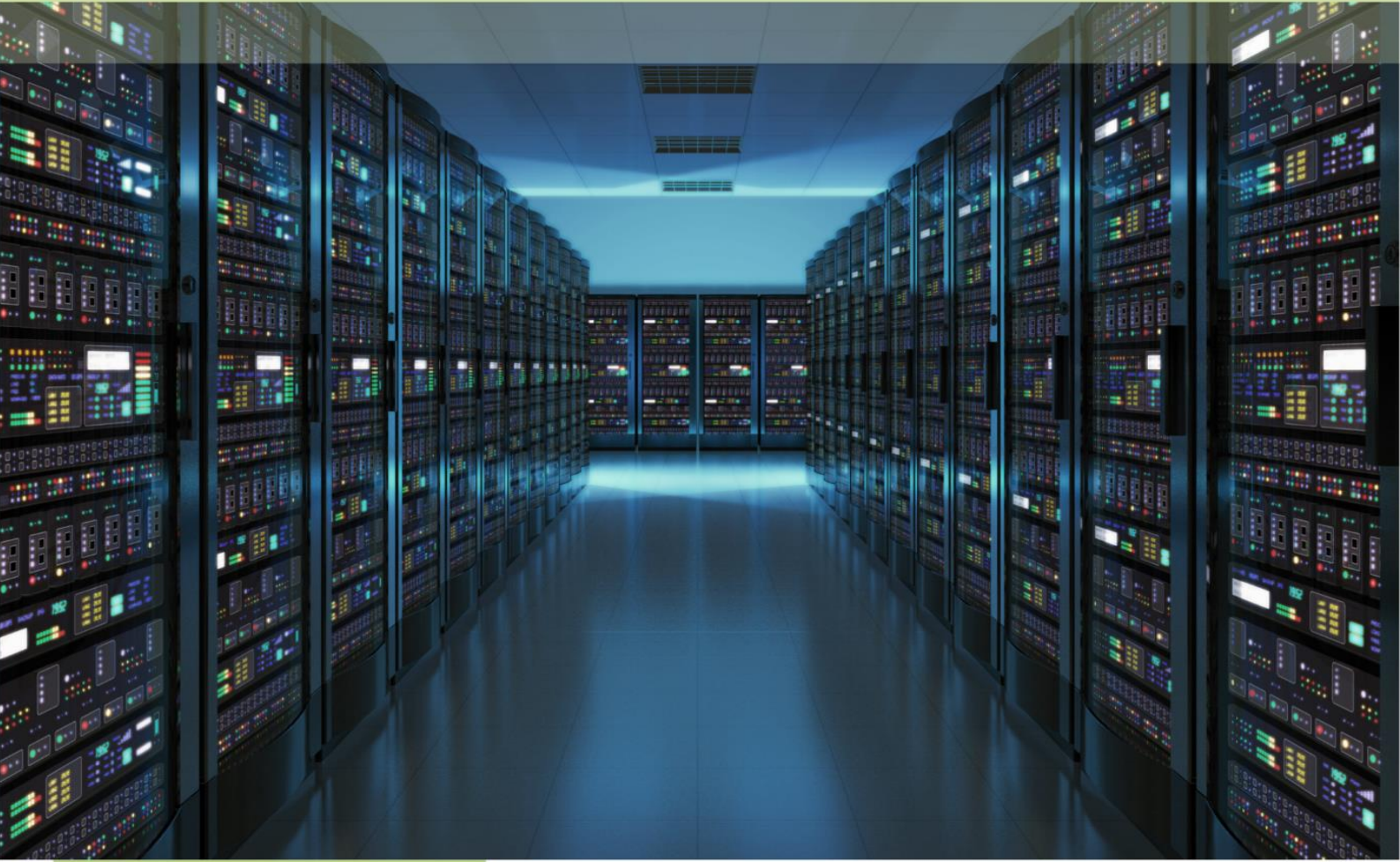
## HIGH VOLTAGE LITHIUM SOLUTION

Our HV range of Lithium backup Solutions are designed for High Voltage UPS & Energy storage Applications and are compatible with all kinds of two Wire (Non-centre tap) and three wire (Centre tap) systems.



Pictorial representation for reference purpose only

# ENERTECT VISION TECHNOLOGIES PVT. LTD.



Global Power Source Group of Companies have dominated the Power Backup Industry since 1988. With 30 plus years of experience and a strong network that spans over 30 countries, we continuously strive to provide high-quality industrial products, solutions, and service around the globe with an assurance of unmatched quality.

We aim to be renowned pioneers while continuously evolving to provide exceptional, innovative and effective solutions. With our new division dedicated for Lithium batteries namely

“ENERTECT VISION TECHNOLOGIES PVT. LTD. (EVTPL)” we look forward to cater the backup industry with the latest technology and high quality of performance as per tailor made customer's requirement.

EVTPL's Lithium battery solutions for UPS& Solar energy storage applications have been developed in house and tested vigorously for these applications. The BMS used in this system is developed locally and in-house in cooperation with Ellithion technology from USA as per the specific design requirement for these application. This BMS monitors, controls and enhances the performance of the battery solution effectively as these packs are used for critical and important applications. EVTPL follows the make-in-India initiative completely and each pack is manufactured in local facility near Mumbai, India with the highest level of quality regulations.

# LITHIUM BATTERY PACK UPS APPLICATIONS

## HVL 192V 60AH

Strict choice for cells:

We ensure the good consistency of the product by choosing the highest grade and quality of cells. The cells are selected based on the capacity, internal resistance and the lowest delta voltage difference.

Full-featured BMS:

We use the advance featured BMS system which focuses on utmost operational and Environmental safety. The BMS not only monitors temperature, voltage and capacity but also can regulate and protect the modules and battery pack with its advance technology.

Rack arrangement:

The battery packs are designed in modular rack system which focuses on optimum foot print in terms of space and weight.

Stable system performance:

The battery pack are time tested to provide stable performance with benefit to the user in terms of return on investment (ROI), space utilization, lower carbon foot print.



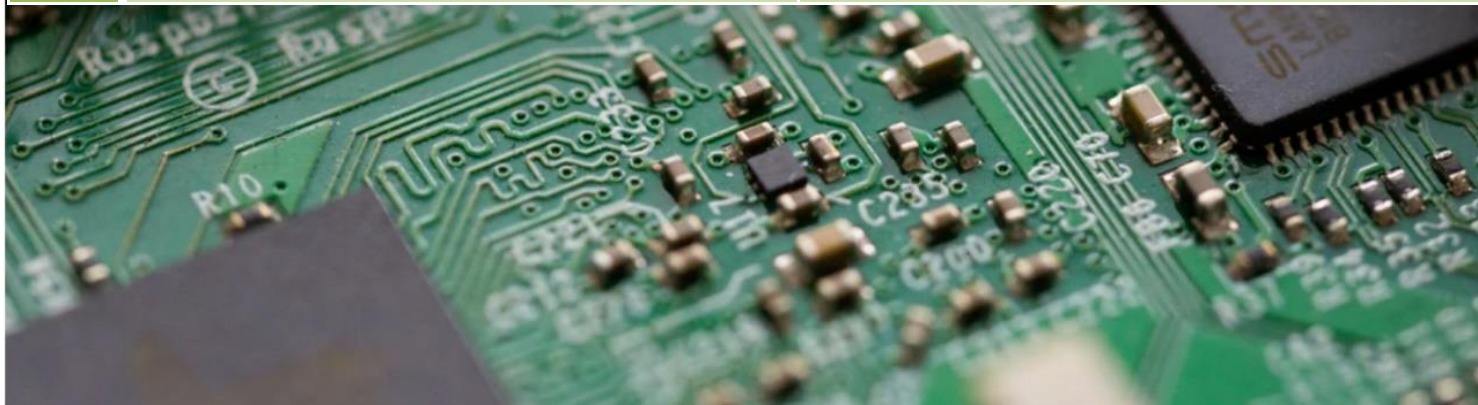
## LITHIUM BATTERY FEATURES

| NO. | FEATURES                         | REMARKS                                                                                                          |
|-----|----------------------------------|------------------------------------------------------------------------------------------------------------------|
| 1.  | <b>Flexibility</b>               | Solutions are customized to match the exact requirement of the application & compatibility with the UPS systems. |
| 2.  | <b>Form Factor</b>               | Our solution ensures the smallest footprint built in customized racks.                                           |
| 3.  | <b>Float service life @25°C</b>  | 8 to 10 years                                                                                                    |
| 4.  | <b>No. of Cycles</b>             | 3000 cycles at 80% DOD. (As per cell chemistry and Charge / Discharge Ratings)                                   |
| 5.  | <b>Rate of Discharge</b>         | High Rate Discharge / Longer Backup solutions                                                                    |
| 6.  | <b>General Usage Temperature</b> | 25°C +/- 2 (recommended for Optimal Performance)                                                                 |
| 7.  | <b>Usage</b>                     | Float / Cyclic depending on Application requirement                                                              |
| 8.  | <b>Weight</b>                    | Smaller foot print as compared to other technologies                                                             |



# BATTERY PACK SPECIFICATIONS

| NO. | PARAMETER                                                           | STANDARD VALUE                                                                                                                                                                                                                                                                                                                                                   |
|-----|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.  | Nominal Voltage                                                     | 192V                                                                                                                                                                                                                                                                                                                                                             |
| 2.  | Typical Capacity<br>(Capacity can vary by +/- 2%)                   | 60Ah                                                                                                                                                                                                                                                                                                                                                             |
| 3.  | Maximum Allowable Discharge Current of Battery Pack                 | 120A (Max. 2C)                                                                                                                                                                                                                                                                                                                                                   |
| 4.  | Discharge cut-off voltage                                           | 168V                                                                                                                                                                                                                                                                                                                                                             |
| 5.  | Charge Voltage<br>(Charge Mode: CC/CV, use special Lithium Charger) | 216V                                                                                                                                                                                                                                                                                                                                                             |
| 6.  | Charge Current                                                      | 6A to 30A or 0.1C to 0.5C                                                                                                                                                                                                                                                                                                                                        |
| 7.  | Allowable Discharge current as per BMS rating                       | 30A                                                                                                                                                                                                                                                                                                                                                              |
| 8.  | Total Available Energy of System                                    | 11.5kwh                                                                                                                                                                                                                                                                                                                                                          |
| 9.  | No. of Modules                                                      | (4 Module+1BPU) per System*1                                                                                                                                                                                                                                                                                                                                     |
| 10. | Modules Capacity                                                    | 48V 60Ah                                                                                                                                                                                                                                                                                                                                                         |
| 11. | Battery configuration (2 Wire or 3 Wire)                            | 2 Wire                                                                                                                                                                                                                                                                                                                                                           |
| 12. | Cell Chemistry                                                      | LFP                                                                                                                                                                                                                                                                                                                                                              |
| 13. | Cell Type                                                           | Cylindrical                                                                                                                                                                                                                                                                                                                                                      |
| 14. | Recommended Operating Temperature                                   | 25°C +/- 2°C                                                                                                                                                                                                                                                                                                                                                     |
| 15. | Total Weight (Approx.)                                              | 320kg per Rack * 1                                                                                                                                                                                                                                                                                                                                               |
| 16. | Size (W*D*H) (+/-1%) (Can be tailor made)                           | (600mm * 800mm * 1704mm) per Rack *1                                                                                                                                                                                                                                                                                                                             |
| 17. | Temperature                                                         | <ul style="list-style-type: none"> <li>Working temperature : <ul style="list-style-type: none"> <li>Charging Temperature: 0~45°C</li> <li>Discharging Temperature: -20~60°C</li> </ul> </li> <li>Storage Temperature: 0~35°C for 6 months</li> </ul>                                                                                                             |
| 18. | Communication Interface                                             | RS485                                                                                                                                                                                                                                                                                                                                                            |
| 19. | Protection Functions                                                | <ul style="list-style-type: none"> <li>Pack over voltage protection</li> <li>Pack under voltage protection</li> <li>Charge over current protection</li> <li>Over &amp; Under Temperature Protection</li> <li>Balance Function</li> <li>Cell over voltage protection</li> <li>Discharge over current protection</li> <li>Cell under voltage protection</li> </ul> |



# ELECTRICAL CHARACTERISTICS OF BATTERY MANAGEMENT SYSTEM

- BMS monitors the total voltage of battery as well as of individual cells which helps users to determine the health and performance of the batteries.
- Measures the amount of current which flows through each battery as this enables the user to optimize the current flow in the system.
- Temperature at pack level is measured to help users prevent thermal runaway and overheating.
- It measures and indicates the battery capacity in percentage, which helps the user to accurately determine the power of battery pack.

## TECHNICAL SPECIFICATIONS

| PARAMETER                                  | FUNCTION                                     | REMARKS                                                 |
|--------------------------------------------|----------------------------------------------|---------------------------------------------------------|
| Voltage                                    | Charge Mode                                  | CC/CV                                                   |
|                                            | Single cell charged balance Voltage          | Available                                               |
| Current                                    | Continuous charge /discharge current         | Available                                               |
| Over voltage protection                    | Overcharge protection voltage                | Available                                               |
|                                            | Overcharge release voltage                   | Available                                               |
| Under voltage protection                   | Under discharge Voltage protection           | Available                                               |
|                                            | Under discharge release voltage              | Available                                               |
| Over current protection                    | Overcharge current protection function       | Available                                               |
|                                            | Over discharge current protection function   | Available                                               |
|                                            | Over discharge current release function      | Cut off Load, release automatically                     |
| Short circuit protection                   | Condition                                    | Exterior Short Circuit                                  |
|                                            | Release Condition                            | Cut off Load, release automatically                     |
| Temperature                                | Overcharge/ Discharge Temperature protection | Available                                               |
| Remote alert notification (Optional & Ons) | Via Internet or Sim cards                    | Subscription & Device charges apply extra as applicable |



### HVL BATTERY PACK SERIES

EVTP's High voltage Lithium battery Series is offered for UPS & Energy Storage Applications.

### WHY EVTP?

EVTP's High voltage Lithium battery Series adopts the leading Lithium technology to ensure the batteries are designed for longer float life, optimum performance, operational safety and environmental safety. The battery pack are designed for each customized application specifically.

### STANDARD WARRANTY

Five Years

### SERVICES AVAILABLE

Technical Support  
Installation  
Commissioning  
Maintenance  
Application Support  
Hardware Support

### OTHER AVAILABLE PRODUCTS

Energy Storage Solutions  
Hybrid Solar Solutions  
Electric Vehicle Solutions

### APPLICATION:

UPS / ESS

### VISIT US AT:



For warranty details, refer to T&C of warranty usage and handling.